Homework 5: System F – Solutions

Problem 1

In each part, M denotes the given term; derivation names are local to that part.

(a)(i) Typing

$$\tau = \mathtt{T} \to (\mathtt{T} \to \mathtt{T}) \to \mathtt{T}$$

The typing derivation is:

$$\frac{\mathcal{D}_{\text{type}} \quad \mathcal{D}_{\text{eval}}}{\bullet;\bullet \vdash M : \mathtt{T} \to (\mathtt{T} \to \mathtt{T}) \to \mathtt{T}}\ \text{T-App}$$

Here $\mathcal{D}_{\text{type}}$ denotes the following derivation tree:

$$\frac{\mathcal{D}_{\text{type-beta}} \quad \bullet \vdash \mathtt{T}}{\begin{array}{c}\bullet;\bullet \vdash (\Lambda\alpha.\Lambda\beta.\Lambda\gamma.\lambda f : \alpha \to \beta \to \gamma.\lambda y : \beta.\lambda x : \alpha.f\,x\,y) \\ \langle \mathtt{T} \to \mathtt{T}\rangle\langle \mathtt{T}\rangle\langle \mathtt{T}\rangle : \\ ((\mathtt{T} \to \mathtt{T}) \to \mathtt{T} \to \mathtt{T}) \to \mathtt{T} \to (\mathtt{T} \to \mathtt{T}) \to \mathtt{T}\end{array}}\ \text{T-TApp}$$

Use $\mathcal{D}_{\text{type-beta}}$ to denote the following derivation tree:

$$\frac{\mathcal{D}_{\text{type-alpha}} \quad \bullet \vdash \mathtt{T}}{\begin{array}{c}\bullet;\bullet \vdash (\Lambda\alpha.\Lambda\beta.\Lambda\gamma.\lambda f : \alpha \to \beta \to \gamma.\lambda y : \beta.\lambda x : \alpha.f\,x\,y) \\ \langle \mathtt{T} \to \mathtt{T}\rangle\langle \mathtt{T}\rangle : \\ \forall\gamma.((\mathtt{T} \to \mathtt{T}) \to \mathtt{T} \to \gamma) \to \mathtt{T} \to (\mathtt{T} \to \mathtt{T}) \to \gamma\end{array}}\ \text{T-TApp}$$

Use $\mathcal{D}_{\text{type-alpha}}$ to denote the following derivation tree:

$$\frac{\mathcal{D}_{\text{poly}} \quad \bullet \vdash \mathtt{T} \to \mathtt{T}}{\begin{array}{c}\bullet;\bullet \vdash (\Lambda\alpha.\Lambda\beta.\Lambda\gamma.\lambda f : \alpha \to \beta \to \gamma.\lambda y : \beta.\lambda x : \alpha.f\,x\,y) \\ \langle \mathtt{T} \to \mathtt{T}\rangle : \\ \forall\beta.\forall\gamma.((\mathtt{T} \to \mathtt{T}) \to \beta \to \gamma) \to \beta \to (\mathtt{T} \to \mathtt{T}) \to \gamma\end{array}}\ \text{T-TApp}$$

Use $\mathcal{D}_{\text{poly}}$ to denote the following derivation tree:

$$\frac{\dfrac{\dfrac{\dfrac{\alpha,\beta,\gamma \vdash \alpha \to \beta \to \gamma \quad \mathcal{D}_{\text{arguments}}}{\alpha,\beta,\gamma;\bullet \vdash \lambda f : \alpha \to \beta \to \gamma.\lambda y : \beta.\lambda x : \alpha.f\,x\,y :\ (\alpha \to \beta \to \gamma) \to \beta \to \alpha \to \gamma}\ \text{T-Abs}}{\alpha,\beta;\bullet \vdash \Lambda\gamma.\lambda f : \alpha \to \beta \to \gamma.\lambda y : \beta.\lambda x : \alpha.f\,x\,y :\ \forall\gamma.(\alpha \to \beta \to \gamma) \to \beta \to \alpha \to \gamma}\ \text{T-TAbs}}{\alpha;\bullet \vdash \Lambda\beta.\Lambda\gamma.\lambda f : \alpha \to \beta \to \gamma.\lambda y : \beta.\lambda x : \alpha.f\,x\,y :\ \forall\beta.\forall\gamma.(\alpha \to \beta \to \gamma) \to \beta \to \alpha \to \gamma}\ \text{T-TAbs}}{\bullet;\bullet \vdash \Lambda\alpha.\Lambda\beta.\Lambda\gamma.\lambda f : \alpha \to \beta \to \gamma.\lambda y : \beta.\lambda x : \alpha.f\,x\,y :\ \forall\alpha.\forall\beta.\forall\gamma.(\alpha \to \beta \to \gamma) \to \beta \to \alpha \to \gamma}\ \text{T-TAbs}$$

Let

$$\Delta = \alpha,\beta,\gamma, \qquad \Gamma = f : \alpha \to \beta \to \gamma, \qquad \Gamma_1 = \Gamma, y : \beta, x : \alpha$$

Use $\mathcal{D}_{\text{arguments}}$ to denote the following derivation tree:

$$\dfrac{\Delta \vdash \beta \quad \dfrac{\Delta \vdash \alpha \quad \mathcal{D}_{\text{body}}}{\Delta; \Gamma, y : \beta \vdash \lambda x : \alpha . f\, x\, y : \alpha \to \gamma}\ \text{T-Abs}}{\Delta; \Gamma \vdash \lambda y : \beta . \lambda x : \alpha . f\, x\, y : \beta \to \alpha \to \gamma}\ \text{T-Abs}$$

Use $\mathcal{D}_{\text{body}}$ to denote the following derivation tree:

$$\dfrac{\mathcal{D}_{\text{inner}} \quad \dfrac{y : \beta \in \Gamma_1}{\Delta; \Gamma_1 \vdash y : \beta}\ \text{T-Var}}{\Delta; \Gamma_1 \vdash f\, x\, y : \gamma}\ \text{T-App}$$

Use $\mathcal{D}_{\text{inner}}$ to denote the following derivation tree:

$$\dfrac{\dfrac{f : \alpha \to \beta \to \gamma \in \Gamma_1}{\Delta; \Gamma_1 \vdash f : \alpha \to \beta \to \gamma}\ \text{T-Var} \quad \dfrac{x : \alpha \in \Gamma_1}{\Delta; \Gamma_1 \vdash x : \alpha}\ \text{T-Var}}{\Delta; \Gamma_1 \vdash f\, x : \beta \to \gamma}\ \text{T-App}$$

Use $\mathcal{D}_{\text{eval}}$ to denote the following derivation tree:

$$\dfrac{\bullet \vdash \mathtt{T} \to \mathtt{T} \quad \dfrac{\bullet \vdash \mathtt{T} \quad \mathcal{D}_{\text{eval-body}}}{\bullet; g : \mathtt{T} \to \mathtt{T} \vdash \lambda z : \mathtt{T} . g\, z : \mathtt{T} \to \mathtt{T}}\ \text{T-Abs}}{\begin{array}{c}\bullet; \bullet \vdash \lambda g : \mathtt{T} \to \mathtt{T} . \lambda z : \mathtt{T} . g\, z : \\ (\mathtt{T} \to \mathtt{T}) \to \mathtt{T} \to \mathtt{T}\end{array}}\ \text{T-Abs}$$

Let $\Gamma_2 = g : \mathtt{T} \to \mathtt{T}, z : \mathtt{T}$. Use $\mathcal{D}_{\text{eval-body}}$ to denote the following derivation tree:

$$\dfrac{\dfrac{g : \mathtt{T} \to \mathtt{T} \in \Gamma_2}{\bullet; \Gamma_2 \vdash g : \mathtt{T} \to \mathtt{T}}\ \text{T-Var} \quad \dfrac{z : \mathtt{T} \in \Gamma_2}{\bullet; \Gamma_2 \vdash z : \mathtt{T}}\ \text{T-Var}}{\bullet; \Gamma_2 \vdash g\, z : \mathtt{T}}\ \text{T-App}$$

Problem 1 (continued)

(a)(ii) Translation

$$|M| = (\lambda f.\lambda y.\lambda x.f\ x\ y)(\lambda g.\lambda z.g\ z)$$

(a)(iii) Reduction

$$\begin{aligned}
M &\longrightarrow_F \big(\Lambda\beta.\Lambda\gamma.\lambda f : (\mathtt{T} \to \mathtt{T}) \to \beta \to \gamma. \\
&\qquad \lambda y : \beta.\lambda x : \mathtt{T} \to \mathtt{T}.f\ x\ y\big)\langle\mathtt{T}\rangle\langle\mathtt{T}\rangle(\lambda g : \mathtt{T} \to \mathtt{T}.\lambda z : \mathtt{T}.g\ z) \\
&\longrightarrow_F \big(\Lambda\gamma.\lambda f : (\mathtt{T} \to \mathtt{T}) \to \mathtt{T} \to \gamma.\lambda y : \mathtt{T}.\lambda x : \mathtt{T} \to \mathtt{T}.f\ x\ y\big) \\
&\qquad \langle\mathtt{T}\rangle(\lambda g : \mathtt{T} \to \mathtt{T}.\lambda z : \mathtt{T}.g\ z) \\
&\longrightarrow_F (\lambda f : (\mathtt{T} \to \mathtt{T}) \to \mathtt{T} \to \mathtt{T}.\lambda y : \mathtt{T}.\lambda x : \mathtt{T} \to \mathtt{T}.f\ x\ y) \\
&\qquad (\lambda g : \mathtt{T} \to \mathtt{T}.\lambda z : \mathtt{T}.g\ z) \\
&\longrightarrow_F \lambda y : \mathtt{T}.\lambda x : \mathtt{T} \to \mathtt{T}.(\lambda g : \mathtt{T} \to \mathtt{T}.\lambda z : \mathtt{T}.g\ z)\ x\ y \\
&\longrightarrow_F \lambda y : \mathtt{T}.\lambda x : \mathtt{T} \to \mathtt{T}.(\lambda z : \mathtt{T}.x\ z)\ y \\
&\longrightarrow_F \lambda y : \mathtt{T}.\lambda x : \mathtt{T} \to \mathtt{T}.x\ y
\end{aligned}$$

$$\begin{aligned}
|M| &= (\lambda f.\lambda y.\lambda x.f\ x\ y)(\lambda g.\lambda z.g\ z) \\
&\longrightarrow_u \lambda y.\lambda x.(\lambda g.\lambda z.g\ z)\ x\ y \\
&\longrightarrow_u \lambda y.\lambda x.(\lambda z.x\ z)\ y \\
&\longrightarrow_u \lambda y.\lambda x.x\ y
\end{aligned}$$

Problem 1 (continued)

(b)(i) Typing

$$\tau = \mathtt{T} \to \mathtt{T} \to \mathtt{T}$$

The typing derivation is:

$$\frac{\mathcal{D}_{\text{type}} \quad \mathcal{D}_{\text{selector}}}{\bullet;\bullet \vdash M : \mathtt{T} \to \mathtt{T} \to \mathtt{T}}\ \text{T-App}$$

Here $\mathcal{D}_{\text{type}}$ denotes the following derivation tree:

$$\frac{\mathcal{D}_{\text{poly}} \quad \bullet \vdash \mathtt{T} \to \mathtt{T} \to \mathtt{T}}{\begin{array}{c}\bullet;\bullet \vdash \big(\Lambda\gamma.\lambda k : (\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to \gamma. \\ k\,(\lambda z : \mathtt{T}.z)(\lambda u : \mathtt{T}.\lambda v : \mathtt{T}.u)\big)\langle \mathtt{T} \to \mathtt{T} \to \mathtt{T}\rangle : \\ \big((\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T})\big) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T})\end{array}}\ \text{T-TApp}$$

Use $\mathcal{D}_{\text{poly}}$ to denote the following derivation tree:

$$\frac{\mathcal{D}_{\text{abs}}}{\begin{array}{c}\bullet;\bullet \vdash \Lambda\gamma.\lambda k : (\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to \gamma. \\ k\,(\lambda z : \mathtt{T}.z)(\lambda u : \mathtt{T}.\lambda v : \mathtt{T}.u) : \\ \forall\gamma.\big((\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to \gamma\big) \to \gamma\end{array}}\ \text{T-TAbs}$$

Let

$$\Gamma = k : (\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to \gamma$$

Use $\mathcal{D}_{\text{abs}}$ to denote the following derivation tree:

$$\frac{\gamma \vdash (\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to \gamma \quad \mathcal{D}_{\text{body}}}{\begin{array}{c}\gamma;\bullet \vdash \lambda k : (\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to \gamma. \\ k\,(\lambda z : \mathtt{T}.z)(\lambda u : \mathtt{T}.\lambda v : \mathtt{T}.u) : \\ \big((\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to \gamma\big) \to \gamma\end{array}}\ \text{T-Abs}$$

Use $\mathcal{D}_{\text{body}}$ to denote the following derivation tree:

$$\frac{\mathcal{D}_{\text{first}} \quad \mathcal{D}_{\text{proj}}}{\gamma;\Gamma \vdash k\,(\lambda z : \mathtt{T}.z)(\lambda u : \mathtt{T}.\lambda v : \mathtt{T}.u) : \gamma}\ \text{T-App}$$

Use $\mathcal{D}_{\text{first}}$ to denote the following derivation tree:

$$\frac{\dfrac{k : (\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to \gamma \in \Gamma}{\gamma;\Gamma \vdash k : (\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to \gamma}\ \text{T-Var} \quad \mathcal{D}_{\text{id}}}{\gamma;\Gamma \vdash k\,(\lambda z : \mathtt{T}.z) : (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to \gamma}\ \text{T-App}$$

Use $\mathcal{D}_{\text{id}}$ to denote the following derivation tree:

$$\frac{\gamma \vdash \mathtt{T} \quad \dfrac{z : \mathtt{T} \in \Gamma, z : \mathtt{T}}{\gamma;\Gamma, z : \mathtt{T} \vdash z : \mathtt{T}}\ \text{T-Var}}{\gamma;\Gamma \vdash \lambda z : \mathtt{T}.z : \mathtt{T} \to \mathtt{T}}\ \text{T-Abs}$$

Use $\mathcal{D}_{\text{proj}}$ to denote the following derivation tree:

$$\frac{\gamma \vdash \mathtt{T} \quad \dfrac{\gamma \vdash \mathtt{T} \quad \dfrac{u : \mathtt{T} \in \Gamma, u : \mathtt{T}, v : \mathtt{T}}{\gamma;\Gamma, u : \mathtt{T}, v : \mathtt{T} \vdash u : \mathtt{T}}\ \text{T-Var}}{\gamma;\Gamma, u : \mathtt{T} \vdash \lambda v : \mathtt{T}.u : \mathtt{T} \to \mathtt{T}}\ \text{T-Abs}}{\gamma;\Gamma \vdash \lambda u : \mathtt{T}.\lambda v : \mathtt{T}.u : \mathtt{T} \to \mathtt{T} \to \mathtt{T}}\ \text{T-Abs}$$

Use $\mathcal{D}_{\text{selector}}$ to denote the following derivation tree:

$$\dfrac{\bullet \vdash \mathtt{T} \to \mathtt{T} \quad \mathcal{D}_{\text{selector}-\text{body}}}{\bullet;\bullet \vdash \lambda f : \mathtt{T} \to \mathtt{T}.\lambda g : \mathtt{T} \to \mathtt{T} \to \mathtt{T}.g : (\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T})}\ \text{T-Abs}$$

Let $\Gamma_1 = f : \mathtt{T} \to \mathtt{T}$. Use $\mathcal{D}_{\text{selector}-\text{body}}$ to denote the following derivation tree:

$$\dfrac{\bullet \vdash \mathtt{T} \to \mathtt{T} \to \mathtt{T} \quad \dfrac{g : \mathtt{T} \to \mathtt{T} \to \mathtt{T} \in \Gamma_1, g : \mathtt{T} \to \mathtt{T} \to \mathtt{T}}{\bullet;\Gamma_1, g : \mathtt{T} \to \mathtt{T} \to \mathtt{T} \vdash g : \mathtt{T} \to \mathtt{T} \to \mathtt{T}}\ \text{T-Var}}{\bullet;\Gamma_1 \vdash \lambda g : \mathtt{T} \to \mathtt{T} \to \mathtt{T}.g : (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T})}\ \text{T-Abs}$$

Problem 1 (continued)

(b)(ii) Translation

$$|M| = (\lambda k.k\,(\lambda z.z)(\lambda u.\lambda v.u))(\lambda f.\lambda g.g)$$

(b)(iii) Reduction

$$\begin{aligned}
M \longrightarrow_F\ & \big(\lambda k : (\mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}) \to (\mathtt{T} \to \mathtt{T} \to \mathtt{T}).\\
& \quad k\,(\lambda z : \mathtt{T}.z)(\lambda u : \mathtt{T}.\lambda v : \mathtt{T}.u)\big)\\
& \quad (\lambda f : \mathtt{T} \to \mathtt{T}.\lambda g : \mathtt{T} \to \mathtt{T} \to \mathtt{T}.g)\\
\longrightarrow_F\ & (\lambda f : \mathtt{T} \to \mathtt{T}.\lambda g : \mathtt{T} \to \mathtt{T} \to \mathtt{T}.g)\\
& \quad (\lambda z : \mathtt{T}.z)(\lambda u : \mathtt{T}.\lambda v : \mathtt{T}.u)\\
\longrightarrow_F\ & (\lambda g : \mathtt{T} \to \mathtt{T} \to \mathtt{T}.g)(\lambda u : \mathtt{T}.\lambda v : \mathtt{T}.u)\\
\longrightarrow_F\ & \lambda u : \mathtt{T}.\lambda v : \mathtt{T}.u
\end{aligned}$$

$$\begin{aligned}
|M| &= (\lambda k.k\,(\lambda z.z)(\lambda u.\lambda v.u))(\lambda f.\lambda g.g)\\
&\longrightarrow_u (\lambda f.\lambda g.g)(\lambda z.z)(\lambda u.\lambda v.u)\\
&\longrightarrow_u (\lambda g.g)(\lambda u.\lambda v.u)\\
&\longrightarrow_u \lambda u.\lambda v.u
\end{aligned}$$

2. Properties of type erasure

(a) No. Take $M = (\Lambda\alpha.\lambda x : \alpha.x)\langle \mathtt{T} \rangle$. We have

$$|M| = \lambda x.x, \qquad M \longrightarrow_F \lambda x : \mathtt{T}.x.$$

The erasure is an untyped normal form, while the source has a type-beta step.

(b) No. Take

$$M = (\Lambda\alpha.\lambda x : \mathtt{T}.x)(\lambda y : \mathtt{T}.y).$$

The outer application has a type abstraction as its function, so no beta rule applies. Neither subterm contains a redex. Thus M is a System F normal form. However,

$$|M| = (\lambda x.x)(\lambda y.y) \longrightarrow_u \lambda y.y,$$

so its erasure is not an untyped normal form. This counterexample is closed and ill typed.

(c) No. Take

$$M = \lambda f : \mathtt{T} \to \mathtt{T}.\lambda x : \mathtt{T}.f\ x,$$
$$N = \lambda f : \mathtt{T}.\lambda x : \mathtt{T}.f\ x.$$

Both terms have the same erasure:

$$|M| = |N| = \lambda f.\lambda x.f\ x.$$

The first term is well typed:

$$\bullet; \bullet \vdash M : (\mathtt{T} \to \mathtt{T}) \to \mathtt{T} \to \mathtt{T}.$$

In N, the variable f has base type $\mathtt{T}$, so the application $f\ x$ cannot be typed by T-App. Thus N is not well typed.